

Covert mmWave Communications With Finite Blocklength Against Spatially Random Wardens

Ruiqian Ma^{1b}, Weiwei Yang^{1b}, Xinrong Guan^{1b}, Xingbo Lu^{1b}, Yi Song^{1b}, and Dechuan Chen^{1b}

Abstract—In this article, we investigate covert millimeter-wave (mmWave) communications with finite blocklength, where a multiantenna transmitter sends covert messages to a legitimate receiver in the presence of spatially random wardens. Both the phase array (PA) and linear frequency diverse array (LFDA) beamforming schemes, which are designed to maximize the antenna gain from the transmitter to the legitimate receiver, are investigated to improve the covert communication performance. First, the novel expressions of covert communication constraint and average effective covert throughput (AECT) are derived for both beamforming schemes. Then, taking into account the constraint of maximal available blocklength, the optimal transmit power and blocklength are determined for maximizing the AECT. Typically, comparing to the benchmark with fixed blocklength, the enhancement of AECT by utilizing the optimized blocklength enlarges as the density of wardens increases. In addition, it is observed that increasing the maximal available blocklength cannot always improve the maximum AECT due to the tradeoff between the transmit power and blocklength. Furthermore, it is

shown that the maximum AECT varies for different directions of the legitimate receiver under both the beamforming schemes, and the transmitter can adaptively choose the PA or LFDA beamforming scheme to improve the covertness performance against spatially random wardens.

Index Terms—Covert communications, finite blocklength, millimeter wave (mmWave), multiantenna, stochastic geometry.

I. INTRODUCTION

A. Background

DUE to the widespread applications of wireless communications, the demand for communication security is growing, especially when the Internet of Things (IoT) has been penetrating into industry as well as our daily life [1], [2], [3]. Most of the existing works on wireless communication security focus on the protection of communication content against eavesdropping, e.g., encryption and physical-layer security [4]. However, in some situations, privacy of communication behavior is also critical, and private messages are required to be conveyed covertly. For example, in the Internet of Battlefield Things, the sensing devices collect and transmit the environmental information in battleground, and the exposure of the transmission between devices and soldiers may disclose the military operations. In this regard, covert communication, which shows the ability to shelter communication itself from being detected, has attracted increasing research attention [5].

As a pioneering work, Bash et al. [6] revealed the fundamental limit of covert communication over additive white Gaussian noise (AWGN) channels. After that, covert communication was extended into various scenarios by considering different practical issues, including noise uncertainty [7], channel uncertainty [8], cooperative jammer [9], age of information [10], relay network [11], and air-ground network [12], [13]. These works mainly considered the case of omnidirectional transmission over conventional radio frequency (RF) bands. By performing covert communication over millimeter-wave (mmWave) bands, a superior performance can be potentially achieved due to the directional beams. Also, the short wavelength of mmWave enables to deploy multiantenna into a compact size device, which greatly expands the applications of mmWave communications [15], [16], [17]. This highlights the importance of covert communication to face security threats in mmWave systems. However, the existing works on covert mmWave communications mainly focused on the situations with single warden [18], [19]. Considering a more general case

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with multiple wardens, the covertness performance may deteriorate dramatically when the wardens locate in the mmWave beams and obtain high antenna gain, of which the covertness framework needs further examinations. On the other hand, for low-energy and delay-sensitive communication scenarios, e.g., IoT and vehicular networks, it is practical to consider the impact of finite blocklength, which is always omitted in the existing works on covert mmWave communication. Against this background, in this article, we investigate the covert mmWave communications with finite blocklength against spatially random wardens, where the related literature and more specific motivations are detained in the following section.

B. Related Works and Motivations

At the early stage of the study, people mainly focused on the enhancement of covert communication performance over RF bands [20], [21], [22], [23], [24]. Specifically, by considering noise uncertainty [20], the effective covert rate were examined under both the conventional and truncated channel inversion power control schemes, in which the location of transmitter can also be hidden. In [21], the detection error probability of the warden was derived and evaluated when the receivers have uncertainty about their channels. It is shown that the covertness performance can be improved when there exists channel estimation error at the warden. In addition to using the uncertainty caused by the randomness of noise and channel, the potential of artificial jamming was exploited for enhancing covertness performance in [22]. In the considered model, a full-duplex receiver was adopted to emit jamming signals, and the power control parameter was optimized to maximize the covert throughput. Furthermore, considering the abundance of communication links in the space, the public links can also be utilized as the cover to hide the existence of the covert links. In [23], covert communication was investigated in a nonorthogonal multiple access system, where the public transmission were utilized to provide a cover for the covert transmission. Considering a single-input-multi-output system with two users (one requires the concealment of message delivery and the other does not), the transmission rate was optimized subject to the covert communication constraint and the quality of service requirement [24]. However, the spectrum resource of conventional RF bands was becoming scarce, and people begin to focus on the higher frequency bands.

Recently, covert mmWave communications have attracted much attention due to its abundance of available spectrum and potential for enhancing covertness performance [18], [19], [25], [26]. In [18], the detection error performance at the warden was investigated by adopting a dual-beam mmWave transmitter, which can transmit message and jamming signals with two independent antenna arrays. Considering that a multiantenna transmitter sends signals to a full-duplex receiver over mmWave channels, the beamforming and jamming vectors were jointly designed in [19] for maximizing the covert rate. Zhang et al. [25] considered that an aerial transmitter adopt a beam sweeping scheme to transmit messages to a ground receiver in the presence of a ground warden. To maximize the covert throughput, the number of beams, the

transmit power, and the height of transmitter were jointly optimized. It is worth mentioning that, these works on covert mmWave communication mainly considered the conventional phase array (PA) of multiantenna, in which the antenna beam has high directivity. In this regard, the exposure of only the user's direction information may cause the failure of secure communication, e.g., the malicious nodes can be deployed in the direction of target receiver to obtain high antenna gain. Differing from the PA of multiantenna, the frequency diverse array can achieve the beam pattern depending on both direction and distance [27]. Considering the covert mmWave communication where one warden locates in the legitimate user's direction, Ma et al. [26] adopted the frequency diverse array beamforming scheme to enhance the covertness performance via suppressing the antenna gain in user's direction (but not the user's location). However, in the case with multiple randomly-distributed wardens, it is still possible that the wardens locate within the beams and thus deteriorates the covertness performance. Although Zheng et al. [28] and Ma et al. [29] investigated the effect of spatially random wardens on covert communications over conventional RF bands, the results cannot be directly extended to covert mmWave communications due to the different properties of mmWave channels, e.g., the line-of-propagation links. Thus, the covert transmission design against spatially random wardens should be reconsidered over mmWave bands.

Most of the above works on covert communications are mainly based on the assumption of infinite blocklength. However, in many practical scenarios, the finite blocklength should be adopted to satisfy the strict power or latency constraint. Furthermore, finite blocklength also means that the warden can only collect limited observations, which naturally causes the uncertainty about the existence of transmission. In this regard, covert communication with finite blocklength has attracted extensive research interest in the literature, e.g., [30], [31], [32], [33], and [34]. Considering that a full-duplex receiver transmits jamming signals, Shu et al. [30] demonstrated that the performance of covert communications can still be improved when the blocklength is finite by adopting fixed jamming power. In [31], the covert communication with random transmit power was studied in the finite blocklength regime, and the optimal transmit power and blocklength maximizing the average effective covert throughput (AECT) were determined. Zhou et al. [32] investigated the covert communication aided by an intelligent reflecting surface, and the transmit power and reflect coefficient were jointly designed subject to the blocklength constraint. In order to hide both of transmitter location and communication behavior, Ma et al. [33] analyzed the covertness performance with finite blocklength under both the conventional and truncated channel inversion power control schemes. Besides, considering a covert mmWave communication system, the beam training duration, training power, and data transmission power that maximize the effective covert throughput were studied in [34]. From these works, we note that the blocklength is a key parameter influencing the covert communication performance, which necessitates blocklength design for covert communications.

As detailed above, performing covert communications on mmWave bands can offer potential for improving the covert communication performance due to the extra spatial degrees of freedom for beamforming design. However, in some specific scenarios, there may exist multiple wardens performing the detection tasks. This implies that the wardens may locate within the mmWave beams, which will deteriorate the communication covertness. Meanwhile, the practical communication scenarios, such as IoT, require finite blocklength, of which the effect on decoding error probability and covertness performance is not negligible. Therefore, in the content of covert communication, the blocklength should be carefully designed by considering its joint effects on the communication covertness and effectiveness. These factors motivate this work to redesign the transmission for achieving enhanced covert mmWave communication against spatially random wardens.

C. Contributions and Notations

In this article, we investigate the joint design of transmit power and blocklength to maximize the AECT in covert mmWave communication, where spatially random wardens attempt to detect the presence of transmission. Our principal contributions and results are summarized as follows.

- 1) Considering the case with spatially random wardens, the tools of stochastic geometry is adopted to describe the location randomness of wardens. Then, the tractable expressions of covert communication constraint are derived under both the PA and linear frequency diverse array (LFDA) beamforming schemes, where the derivations are operated based on the line-of-propagation characteristic of mmWave channels. Moreover, the expression of the AECT, as a function of transmit power, blocklength, antenna number, and warden's density, is proposed to evaluate the covert mmWave performance for both the beamforming schemes.
- 2) To facilitate the covert mmWave transmission design, the optimization problem to maximize the AECT is formulated, of which the optimal transmit power and blocklength are derived for both the PA and LFDA beamforming schemes. Our analysis demonstrates that there exists a nontrivial tradeoff between the transmit power and blocklength for improving the covert communication performance. Besides, it is revealed that the covert communication performance deteriorates as the density of wardens increases.
- 3) The numerical results are presented to gain more insights. Specifically, for both the beamforming schemes, the AECT can be improved by adopting our proposed optimal blocklength comparing to the fixed blocklength case, and the gap becomes large as the density of wardens increases. Besides, increasing the maximal available blocklength cannot always improve the maximum AECT, but we can still adopt a larger number of antennas to improve it. Furthermore, the maximum AECT is related to the orientation of the legitimate receiver relative to the antenna array, and the transmitter can adaptively choose the PA or LFDA

TABLE I
LIST OF MAIN SYMBOLS

Symbol	Description
N	Number of channel uses in one block
M	Number of antennas at Alice
λ_w	Density of Willie
Φ_w	Set of Willie
$\mathbf{h}_{ab}(\mathbf{h}_{aw_l})$	Channel coefficient from Alice to Bob (l -th Willie)
$N_{\max}$	Maximum available blocklength
P_a	Transmit power at Alice
$d_{ab}(d_{aw_l})$	Distance from Alice to Bob (l -th Willie)
$\mathbf{a}_P(\mathbf{a}_L)$	Array steering vector for the PA (LFDA) beamforming scheme
$a_b(a_{w_l})$	Path gain from Alice to Bob (l -th Willie)
$\Psi_m^P(\Psi_m^L)$	Phase difference of the m -th antenna
ξ_{w_l}	Total detection error probability at l -th Willie
σ^2	Variance of AWGN
γ_b	Received signal to noise ratio (SNR) at Bob
R	Channel coding rate
δ	Decoding error probability
ε	Covertness tolerance
$\mathcal{V}_T$	Total variation distance
$\mathcal{D}$	Kullback-Leibler (KL) divergence
$\Omega_P(\Omega_L)$	Average channel gain at the malicious Willie
$G_P(G_L)$	Antenna gain

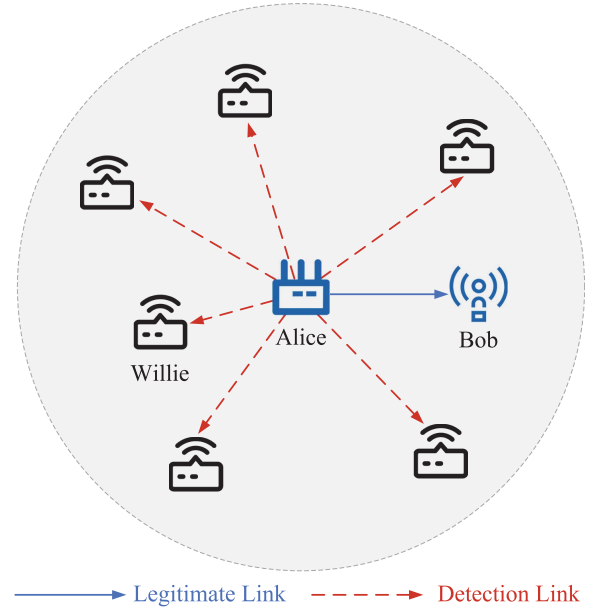


Fig. 1. System model.

beamforming scheme according to the direction of the legitimate receiver to further enhance the covert mmWave communication performance.

Notations: We use the lowercase boldface letters for vectors. For a given vector $\mathbf{x}$, $\mathbf{x}[i]$ denotes the i th elements of $\mathbf{x}$, and $\mathbf{x}^H$ denotes the conjugate transpose of $\mathbf{x}$. The expectation operator is denoted by $\mathbb{E}(\cdot)$, while $\lfloor \cdot \rfloor$ denotes the floor function. $\mathcal{CN}(0, \sigma^2)$ denotes the circularly symmetric complex Gaussian distribution with zero mean and variance σ^2 . The main symbols used in this article are listed in Table I.

II. SYSTEM MODEL

A. Network Description

As shown in Fig. 1, we consider a covert mmWave communication system. Specifically, a multiantenna transmitter

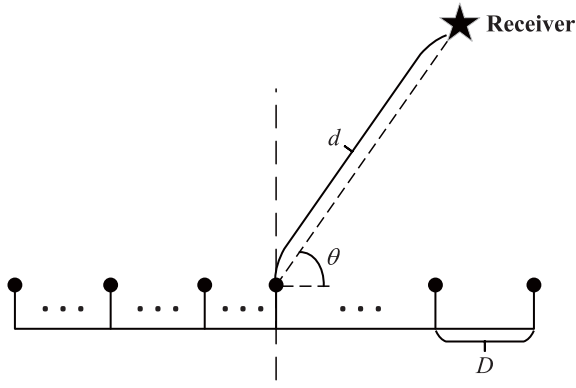


Fig. 2. Illustration of ULA with the reference point at its geometric center.

(Alice) desires to send the privacy information to the receiver (Bob) in the presence of spatially random wardens (Willies), which attempt to detect whether Alice transmits or not. We assume that the finite blocklength with N channel uses and the Gaussian codebooks are adopted at Alice [31]. Alice is equipped with M antennas, and Bob and Willies are all equipped with single antenna. Besides, to characterize the randomness of Willies' locations, the spatially random Willies are modeled as a homogeneous PPP Φ_w with density λ_w [35].

In this work, the channel coefficients from Alice to Bob and the l th Willie are denoted by $\mathbf{h}_{ab}$ and $\mathbf{h}_{awl}$, respectively. Following the 3GPP model, the line-of-sight (LOS) and non-LOS (NLOS) mmWave channels can be modeled as a universal expression by considering different path losses [14]. Then, as suggested by [15], [16], [17], and [26], we adopt the single path model considering the dominant path due to the highly directivity and sparsity of mmWave channels, i.e.,

$$\mathbf{h}_{ai} = \frac{a_i \mathbf{a}(\theta_i)}{d_{ai}^{\alpha/2}} \quad (1)$$

where the subscript $i \in \{b, w_l\}$ denotes Bob or the l th Willie, d_{ai} denotes the distance from Alice to the receiver, α denotes the path-loss exponent with reference distance equaling to 1 m, θ_i is the normalized direction of the path to the receiver, $\mathbf{a}(\theta_i)$ denotes the array steering vector, and a_i is the path gain that follows circularly symmetric complex Gaussian distribution, i.e., $a_i \sim \mathcal{CN}(0, 1)$, which captures that the dominant path can be LOS or NLOS. Specifically, we assume the quasi-static fading channels, which means that a_i remains constant over one block of N channel uses [36].

B. Antenna Array at Alice

As shown in Fig. 2, the uniform linear array (ULA) is adopted at Alice, and both the PA and LFDA beamforming schemes are considered as follows.

1) *PA Beamforming Scheme*: For the PA beamforming scheme, the frequency use at different antenna elements are the same. Set the reference point at the geometric center of array, and the array steering vector at direction θ_i , $i \in \{b, w_l\}$ can be given by [37]

$$\mathbf{a}_P(\theta_i) = \left[e^{j\Psi_0^P(\theta_i)}, \dots, e^{j\Psi_m^P(\theta_i)}, \dots, e^{j\Psi_{M-1}^P(\theta_i)} \right] \quad (2)$$

where $\Psi_m^P(\theta_i)$ is the phase difference of the m th antenna element relative to the reference point, and it can be given by

$$\Psi_m^P(\theta_i) = \left(m - \frac{M-1}{2} \right) \frac{2\pi f_c D \cos(\theta_i)}{c} \quad (3)$$

where $m = 0, 1, \dots, M-1$ denotes the index of antenna elements, D is the interval of contiguous antenna elements, c denotes the speed of light, and f_c is the carrier frequency.

Then, the channel coefficient from Alice to Bob and the l th Willie for the PA beamforming scheme can be obtained by substituting (2) into (1), i.e.,

$$\mathbf{h}_{P,ai} = \frac{a_i \mathbf{a}_P(\theta_i)}{d_{ai}^{\alpha/2}}. \quad (4)$$

2) *LFDA Beamforming Scheme*: Differing from the PA beamforming scheme, the transmission frequency linearly increases from one antenna element to the next under the LFDA beamforming scheme [27]. Accordingly, the frequency allocated to the m th antenna element can be given by

$$f_m = f_c + \left(m - \frac{M-1}{2} \right) \Delta f \quad (5)$$

where Δf denotes the frequency increment. Similarly, we set the reference point at the geometric center of the array. Then, considering that the distance between antenna array and the receiver is always much larger than the antenna element interval, the distance from the m th antenna element to the receiver can be approximated as [38], [39]

$$d_{ai}^m = d_{ai} - \left(m - \frac{M-1}{2} \right) D \cos(\theta_i) \quad (6)$$

where d_{ai} denotes the distance from the reference point to the receiver, i.e., the distance from Alice to the receiver.

It is known that the phase of transmit signal is related to both the carrier frequency and transmission distance. Thus, from (6), the phase shift between two antenna elements depends on both θ_i and d_{ai} due to the existence of frequency increment under the LFDA beamforming scheme. Denoting the array steering vector for the LFDA beamforming scheme as $\mathbf{a}_L(\theta_i, d_{ai})$, the expression of which can be given by

$$\mathbf{a}_L(\theta_i, d_{ai}) = \left[e^{j\Psi_0^L(\theta_i, d_{ai})}, \dots, e^{j\Psi_m^L(\theta_i, d_{ai})}, \dots, e^{j\Psi_{M-1}^L(\theta_i, d_{ai})} \right] \quad (7)$$

where the phase shift of the m th antenna element relative to the reference point, i.e., $\Psi_m^L(\theta_i, d_{ai})$, can be derived as

$$\begin{aligned}
\Psi_m^L(\theta_i, d_{ai}) &= 2\pi f_m \frac{d_{ai}^m}{c} - 2\pi f_c \frac{d_{ai}}{c} \\
&= 2\pi \left(f_c + \left(m - \frac{M-1}{2} \right) \Delta f \right) \\
&\quad \times \frac{d_{ai} - \left(m - \frac{M-1}{2} \right) D \cos(\theta_i)}{c} - \frac{2\pi f_c d_{ai}}{c} \\
&= \frac{2\pi}{c} \left(-f_c \left(m - \frac{M-1}{2} \right) D \cos(\theta_i) \right. \\
&\quad \left. + d_{ai} \left(m - \frac{M-1}{2} \right) \Delta f \right. \\
&\quad \left. - \left(m - \frac{M-1}{2} \right)^2 \Delta f D \cos(\theta_i) \right) \\
&\stackrel{a}{\approx} \frac{2\pi}{c} \left(-f_c \left(m - \frac{M-1}{2} \right) D \cos(\theta_i) \right. \\
&\quad \left. + d_{ai} \left(m - \frac{M-1}{2} \right) \Delta f \right) \quad (8)
\end{aligned}$$

where step (a) is due to the fact that $M\Delta f \ll f_c$ and $MD \ll d_{ai}$ [39], [40], [41].

Thus, the channel coefficient from Alice to Bob and the l th Willie for the LFDA beamforming scheme can be given by

$$\mathbf{h}_{L,ai} = \frac{a_i \mathbf{a}_L(\theta_i, d_{ai})}{d_{ai}^{\alpha/2}}. \quad (9)$$

C. Detection at Willies

In the considered system, there exist spatially random Willies detecting the presence of transmission from Alice. The noncolluding Willies, which perform the detection tasks separately [28], is considered. For an arbitrary Willie $w_l \in \Phi_w$, it needs to distinguish the following hypotheses:

$$\begin{cases} \mathcal{H}_0 : \mathbf{y}_{w_l}[i] = \mathbf{n}_{w_l}[i], \\ \mathcal{H}_1 : \mathbf{y}_{w_l}[i] = \sqrt{P_a} \mathbf{h}_{j,aw_l} \mathbf{w}_j \mathbf{x}[i] + \mathbf{n}_{w_l}[i] \end{cases} \quad (10)$$

where P_a is the transmit power at Alice, $\mathbf{w}_j$ denotes the beamforming vector, $\mathbf{y}_{w_l}[i]$ is the received signal at Willie w_l for the i th channel use, $\mathbf{x}[i] \sim \mathcal{CN}(0, 1)$ is the Gaussian codeword, $\mathbf{n}_{w_l}[i] \sim \mathcal{CN}(0, \sigma^2)$ denotes the AWGN, $i = 1, 2, \dots, N$ is the index of channel uses, and the subscript $j \in \{P, L\}$ denotes the PA or LFDA beamforming scheme.

Considering equal prior probabilities, the total detection error probability at the l th Willie w_l is defined as [6], [31]

$$\xi_{w_l} = P_{w_l}^{\text{FA}} + P_{w_l}^{\text{MD}} \quad (11)$$

where $P_{w_l}^{\text{FA}}$ is the false alarm probability that Willie w_l accepts $\mathcal{H}_1$ when $\mathcal{H}_0$ is true, and $P_{w_l}^{\text{MD}}$ denotes the miss detection probability that w_l accepts $\mathcal{H}_0$ when $\mathcal{H}_1$ is true. Naturally, w_l wants to detect the transmission from Alice with the minimum total detection error probability $\xi_{w_l}^*$. According to [6], [7], and [42], the communication is deemed covert for Willie w_l by satisfying the covert communication constraint $\xi_{w_l}^* \geq 1 - \varepsilon$, where the tolerance ε is a predetermined value that represents the covertness level. The constraint $\xi_{w_l}^* \geq 1 - \varepsilon$ means that the detection behavior of Willie w_l is close to a random guess.

Due to the presence of multiple Willies, the communication is deemed covert only when the minimum total detection error

probability of each Willie is lower bounded by $1 - \varepsilon$. Therefore, the covert communication constraint can be expressed as

$$\min_{w_l \in \Phi_w} \xi_{w_l}^* \geq 1 - \varepsilon \quad (12)$$

where Φ_w denotes the set of Willies. Remarkably, (12) can guarantee that each Willie cannot effectively determine the presence of transmission.

D. Transmission From Alice to Bob

In order to maximize the antenna gain from Alice to Bob, the beamforming vectors for the PA and LFDA beamforming schemes are designed as $\mathbf{w}_P = (1/\sqrt{M})\mathbf{a}_P^H(\theta_b)$ and $\mathbf{w}_L = (1/\sqrt{M})\mathbf{a}_L^H(\theta_b, d_{ab})$, respectively. Then, the received SNR at Bob can be given by

$$\gamma_b = \frac{P_a M |a_b|^2 d_{ab}^{-\alpha}}{\sigma^2} \quad (13)$$

where σ^2 denotes the noise power.

Due to the finite blocklength, the decoding error cannot be ignored. Thus, the effective throughput can be given by [30]

$$\eta = NR(1 - \delta) \quad (14)$$

where R is the channel coding rate, and δ is the decoding error probability. For a given R , the expression of δ can be written as [43]

$$\delta = Q\left(\frac{\ln 2 \sqrt{N} (\log_2(1 + \gamma_b) - R)}{\sqrt{1 - (\gamma_b + 1)^{-2}}}\right) \quad (15)$$

where $Q(x) = \int_x^\infty (1/\sqrt{2\pi})e^{-(t^2/2)}dt$ denotes the Q -function.

Note that there exists a random variable a_b in (13), the average effective throughput is adopted in this article, i.e.,

$$\bar{\eta} = NR(1 - \mathbb{E}(\delta)). \quad (16)$$

III. COVERT COMMUNICATIONS FOR PHASE ARRAY BEAMFORMING SCHEME

In this section, considering the PA beamforming scheme at Alice, we first derive a tractable expression of covert communication constraint. Then, the transmit power and blocklength are jointly optimized to maximize the AECT.

A. Covertness Criteria

For robustness, we consider the worst case that Willies can obtain the complete information on the channel from Alice via the training signals. Specifically, during the beam training phase, Alice sequentially scans the transmit antennas to send pilot signals, and Bob conveys back the index of his quantized beamforming vector [44]. Then, Willies may estimate the channel by analyzing the training signals. This indicates that Willies can perform an optimal test, and the minimum total detection error probability can be calculated by [6]

$$\min_{w_l \in \Phi_w} \xi_{w_l}^* = 1 - \max_{w_l \in \Phi_w} \mathcal{V}_T(\mathbb{P}_{1,w_l}^P, \mathbb{P}_{0,w_l}^P) \quad (17)$$

where $\mathbb{P}_{1,w_l}^P$ and $\mathbb{P}_{0,w_l}^P$ are the probability distributions of received signals at Willie w_l under events $\mathcal{H}_1$ and $\mathcal{H}_0$, respectively, and $\mathcal{V}_T(\mathbb{P}_{1,w_l}^P, \mathbb{P}_{0,w_l}^P)$ is the total variation distance between $\mathbb{P}_{1,w_l}^P$ and $\mathbb{P}_{0,w_l}^P$.

As discussed in [6] and [13], the total variation metric is unwieldy for the products of probability measures, which are used in the further analysis. Thus, we use Pinsker's inequality to obtain a tractable upper bound as

$$\mathcal{V}_T(\mathbb{P}_{1,w_l}^P, \mathbb{P}_{0,w_l}^P) \leq \sqrt{\frac{1}{2} \mathcal{D}(\mathbb{P}_{1,w_l}^P \| \mathbb{P}_{0,w_l}^P)} \quad (18)$$

where $\mathcal{D}(\mathbb{P}_{1,w_l}^P \| \mathbb{P}_{0,w_l}^P)$ is the KL divergence from $\mathbb{P}_{1,w_l}^P$ to $\mathbb{P}_{0,w_l}^P$, and its expression is derived as (19), shown at the bottom of the page. Accordingly, after some simple algebraic manipulations, we know that (12) can be guaranteed by satisfying a stricter KL divergence constraint $\max_{w_l \in \Phi_w} \sqrt{(1/2) \mathcal{D}(\mathbb{P}_{1,w_l}^P \| \mathbb{P}_{0,w_l}^P)} \leq \varepsilon$, which results in a robust transmission scheme.

Notably, from the perspective of Alice, it is difficult to obtain the instantaneous channel state information of Willie (i.e., $\mathbf{h}_{P,aw_l}$), especially when Willies keep silent. Therefore, the average KL divergence is adopted to measure the covert-ness performance, which can be written as

$$\mathbb{E} \left(\max_{w_l \in \Phi_w} \sqrt{\frac{1}{2} \mathcal{D}(\mathbb{P}_{1,w_l}^P \| \mathbb{P}_{0,w_l}^P)} \right) \leq \varepsilon. \quad (20)$$

Then, we derive the covert communication constraint in the following theorem.

Theorem 1: For the PA beamforming scheme, the covert communication constraint can be given by

$$\frac{P_a \sqrt{N} \Omega_P}{2\sigma^2} \leq \varepsilon \quad (21)$$

where the average channel gain at the most malicious Willie, i.e., Ω_P , can be expressed as

$$\begin{aligned} \Omega_P &= \frac{\pi \lambda_w}{T} \sum_{j=1}^T \exp \left(-\frac{\lambda_w \tan^{-\frac{2}{\alpha}}(u_j) \Xi_P(u_j)}{\alpha} \right) \frac{\sqrt{u_j (\frac{\pi}{2} - u_j)}}{\cos^2(u_j)} \\ &\quad \times \left(\frac{\pi}{T\alpha} \sum_{i=1}^T e^{-\frac{\tan(u_j)}{G_P(v_i)}} \sqrt{v_i(2\pi - v_i)} + \frac{2 \tan^{-\frac{2}{\alpha}}(u_j) \Xi_P(u_j)}{\alpha^2} \right) \end{aligned} \quad (22)$$

where T is an accuracy and complexity trade-off parameter of Gaussian–Chebyshev quadrature, $v_i = \pi(1 + \cos((2i-1)\pi/2T))$, $u_j = (\pi/4)(1 + \cos((2j-1)\pi/2T))$, the antenna gain $G_P(\cdot)$ can be calculated by

$$G_P(x) = \frac{\sin^2 \left(\frac{M\pi Df_c}{c} (\cos(x) - \cos(\theta_b)) \right)}{M \sin^2 \left(\frac{\pi Df_c}{c} (\cos(x) - \cos(\theta_b)) \right)} \quad (23)$$

and $\Xi_P(\cdot)$ is given by

$$\Xi_P(x) = \frac{\pi}{T} \sum_{i=1}^T \Gamma \left(\frac{2}{\alpha}, \frac{\tan^{-\frac{2}{\alpha}}(x)}{G_P(v_i)} \right) (G_P(v_i))^{\frac{2}{\alpha}} \sqrt{v_i(2\pi - v_i)} \quad (24)$$

where $\Gamma(x, y) = \int_y^\infty e^{-t} t^{x-1} dt$ is the upper incomplete Gamma function [45, eq. (8.350.2)].

Proof: Refer to Appendix A. ■

Remark 1: From (21), it is clear that the increasing of Willie's density λ_w causes the tighter covert communication constraint. This indicates that the covertness performance deteriorates as λ_w increases. To guarantee the communication covertness, Alice can reduce the transmit power and blocklength to degrade the quality of Willies' observation samples.

B. Optimization of Average Effective Covert Throughput

In this work, the AECT is adopted to measure the performance of covert communications [29], [31], [33]. Specifically, the metric AECT is defined as the average amount of information that can be reliably transmitted subject to the covertness constraint, and the larger AECT means the better effectiveness performance while guaranteeing covertness. Based on (16), the AECT for the PA beamforming scheme can be expressed as

$$\bar{\eta}_P = NR(1 - \mathbb{E}(\delta)), \quad \text{s.t. (21)} \quad (25)$$

where the detection error probability δ is given by (15). Note that δ is difficult to be calculated directly due to the existence of Q -function, and thus the linear approximation of

$$\begin{aligned} \mathcal{D}(\mathbb{P}_{1,w_l}^P \| \mathbb{P}_{0,w_l}^P) &= \int_{\mathbf{x}} f(\mathbf{x}|H_1) \ln \frac{f(\mathbf{x}|H_1)}{f(\mathbf{x}|H_0)} d\mathbf{x} \\ &= \int_{-\infty}^{\infty} \prod_{i=1}^N \frac{\exp \left(-\frac{|\mathbf{x}|^2}{P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2 + \sigma^2} \right)}{\pi (P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2 + \sigma^2)} \ln \left(\prod_{i=1}^N \frac{\pi \sigma^2 \exp \left(-\frac{|\mathbf{x}|^2}{P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2 + \sigma^2} \right)}{\pi (P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2 + \sigma^2) \exp \left(-\frac{x^2}{\sigma^2} \right)} \right) d|\mathbf{x}|^2 \\ &= N \left[\ln \frac{\sigma^2}{P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2 + \sigma^2} + \left(\frac{1}{\sigma^2} - \frac{1}{P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2 + \sigma^2} \right) \int_{-\infty}^{\infty} \frac{|\mathbf{x}|^2 \exp \left(-\frac{|\mathbf{x}|^2}{P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2 + \sigma^2} \right)}{\pi (P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2 + \sigma^2)} d|\mathbf{x}|^2 \right] \\ &= N \left(\frac{P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2}{\sigma^2} - \ln \left(1 + \frac{P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2}{\sigma^2} \right) \right). \end{aligned} \quad (19)$$

δ is adopted [46], i.e.,

$$\delta \approx \begin{cases} 1, & \gamma_b < \gamma_0 - \frac{\beta_b}{2\sqrt{N}}, \\ \frac{1}{2} - \frac{\sqrt{N}}{\beta_b}(\gamma_b - \gamma_0), & \gamma_0 - \frac{\beta_b}{2\sqrt{N}} \leq \gamma_b \leq \gamma_0 + \frac{\beta_b}{2\sqrt{N}}, \\ 0, & \gamma_b > \gamma_0 + \frac{\beta_b}{2\sqrt{N}} \end{cases} \quad (26)$$

where $\beta_b = \sqrt{2\pi(2^{2R} - 1)}$, and $\gamma_0 = 2^R - 1$. Then, $\mathbb{E}(\delta)$ can be approximated as (27), shown at the bottom of the page, where the probability distribution function (PDF) and cumulative distribution function (CDF) of γ_b are, respectively, given by

$$f_{\gamma_b}(x) = \begin{cases} \frac{\sigma^2 d_{ab}^\alpha}{P_a M} e^{-\frac{x\sigma^2 d_{ab}^\alpha}{P_a M}}, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (28)$$

and

$$F_{\gamma_b}(x) = \begin{cases} 1 - e^{-\frac{x\sigma^2 d_{ab}^\alpha}{P_a M}}, & x \geq 0 \\ 0, & x < 0. \end{cases} \quad (29)$$

Considering that the blocklength is limited by a maximum value $N_{\max}$, the optimization problem of the transmit power and blocklength to maximize the AECT can be formulated as

$$\begin{aligned} & \max_{N, P_a} \bar{\eta}_P, \\ & \text{s.t. (21), } N \leq N_{\max}, N \in \mathbb{N}^+ \end{aligned} \quad (30)$$

where $\mathbb{N}^+$ denotes the set of positive integers.

To tackle (30), we first determine the optimal transmit power $P_{a,P}^+$ for a given blocklength N . Specifically, it is clear that $\bar{\eta}_P$ is an increasing function of P_a and N . Thus, for a given N , the optimal transmit power $P_{a,P}^+$ should be its maximal available value subject to the covert communication constraint, i.e.,

$$P_{a,P}^+ = \frac{2\varepsilon\sigma^2}{\sqrt{N}\Omega_P} \quad (31)$$

where Ω_P is given by (22). Then, we can derive the optimal transmit power and blocklength in the following theorem.

Theorem 2: Under the PA beamforming scheme, the optimal transmit power and blocklength maximizing the AECT, i.e., the solution of (30), can be given by

$$P_{a,P}^* = \frac{2\varepsilon\sigma^2}{\sqrt{N_P^*}\Omega_P} \quad (32)$$

and

$$N_P^* = \begin{cases} \left\lfloor \left(\frac{\beta_b}{2\gamma_0} \right)^2 \right\rfloor, & N_P^0 \leq \left(\frac{\beta_b}{2\gamma_0} \right)^2 \\ \min(N_{\max}, \lfloor N_P^0 \rfloor), & N_P^0 > \left(\frac{\beta_b}{2\gamma_0} \right)^2 \end{cases} \quad (33)$$

where $N_P^0 = (4M\varepsilon/(\gamma_0\Omega_P d_{ab}^\alpha))^2$.

Proof: Refer to Appendix B. ■

Remark 2: Based on (32) and (33), we know that the average channel gain at the most malicious Willie, i.e., Ω_P , is a significant factor influencing the AECT. From (23), we find that, for a given M , the antenna gain under the PA beamforming scheme is only related to θ , and is maximized at the direction $\theta = \theta_b$. Besides, due to the existence of $\cos(\theta)$ in (23), we find that the decay rates of antenna gain are various for different θ_b when θ varies from θ_b to $\theta_b + \Delta\theta$, where $\Delta\theta$ is a sufficiently small angle. For example, the absolute value of $\cos(90^\circ + \Delta\theta) - \cos(90^\circ)$ is larger than the absolute value of $\cos(0^\circ + \Delta\theta) - \cos(0^\circ)$ due to $|\partial \cos(\theta)/\partial \theta|_{\theta=90^\circ} > |\partial \cos(\theta)/\partial \theta|_{\theta=0^\circ}$. This indicates that the beam-width first decreases and then increases as θ_b increases from 0° to 180° . Thus, Ω_P initially decreases and then enlarges as θ_b increases from 0° to 180° , since the probability that Willies locate within the beam varies. Most importantly, the maximum AECT first increases and then decreases as θ_b enlarges from 0° to 180° , i.e., the covert communication performance for the PA beamforming scheme varies with θ_b .

IV. COVERT COMMUNICATIONS FOR LINEAR FREQUENCY DIVERSE ARRAY BEAMFORMING SCHEME

In this section, we consider that the LFDA beamforming scheme is adopted at Alice. First, we derive a tractable covert communication constraint for the LFDA beamforming scheme. Then, the transmit power and blocklength are jointly optimized to maximize the AECT.

A. Covertiness Criteria

Similar to the PA beamforming scheme in Section III-A, the average KL divergence constraint is adopted, and the covert communication constraint can be expressed as

$$\mathbb{E} \left(\max_{w_l \in \Phi_w} \sqrt{\frac{1}{2} \mathcal{D}(\mathbb{P}_{1,w_l}^L \| \mathbb{P}_{0,w_l}^L)} \right) \leq \varepsilon \quad (34)$$

$$\begin{aligned} \mathbb{E}(\delta) &= \int_0^\infty \mathcal{Q} \left(\frac{\ln 2\sqrt{N}(\log_2(1+x) - R)}{\sqrt{1 - (x+1)^{-2}}} \right) f_{\gamma_b}(x) dx \\ &\approx F_{\gamma_b} \left(\gamma_0 - \frac{\beta_b}{2\sqrt{N}} \right) + \left(\frac{1}{2} + \frac{\gamma_0\sqrt{N}}{\beta_b} - \frac{P_a M \sqrt{N}}{\sigma^2 \beta_b d_{ab}^\alpha} \right) \left(F_{\gamma_b} \left(\gamma_0 + \frac{\beta_b}{2\sqrt{N}} \right) - F_{\gamma_b} \left(\gamma_0 - \frac{\beta_b}{2\sqrt{N}} \right) \right) \\ &\quad - \left(-1 + \left(\frac{\gamma_0\sqrt{N}}{\beta_b} + \frac{1}{2} \right) F_{\gamma_b} \left(\gamma_0 + \frac{\beta_b}{2\sqrt{N}} \right) - \left(\frac{\gamma_0\sqrt{N}}{\beta_b} - \frac{1}{2} \right) F_{\gamma_b} \left(\gamma_0 - \frac{\beta_b}{2\sqrt{N}} \right) \right) \\ &= 1 - \frac{P_a M \sqrt{N}}{\sigma^2 \beta_b d_{ab}^\alpha} \left(F_{\gamma_b} \left(\gamma_0 + \frac{\beta_b}{2\sqrt{N}} \right) - F_{\gamma_b} \left(\gamma_0 - \frac{\beta_b}{2\sqrt{N}} \right) \right) \end{aligned} \quad (27)$$

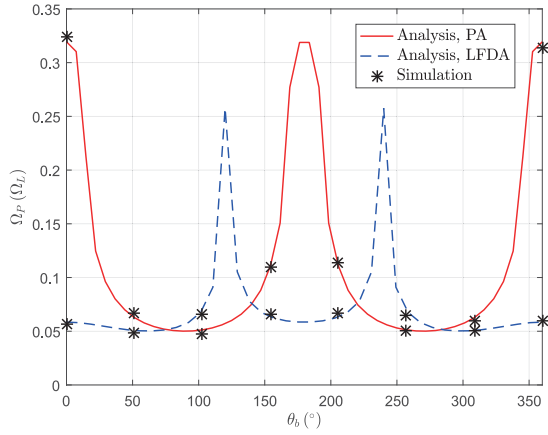


Fig. 3. Average channel gain at the most malicious Willie versus the direction of Bob, where $\lambda_w = 5 \times 10^{-3}$, $\alpha = 2.2$, $M = 32$, and $\Delta f = 1$ MHz.

where $\mathbb{P}_{1,w_l}^L$ and $\mathbb{P}_{0,w_l}^L$ denote the probability distribution of the observations at the l th Willie w_l for the LFDA beamforming scheme under events $\mathcal{H}_1$ and $\mathcal{H}_0$, respectively. Then, the covert communication constraint can be further derived in the following theorem.

Theorem 3: For the LFDA beamforming scheme, the covert communication constraint can be given by

$$\frac{P_a \sqrt{N} \Omega_L}{2\sigma^2} \leq \varepsilon \quad (35)$$

where the average channel gain at the most malicious Willie, i.e., Ω_L , can be expressed as (36), shown at the bottom of the page, $v_i = \pi(1 + \cos((2i-1)\pi/2T))$, $z_j = [(3\pi)/8] + (\pi/8)\cos((2j-1)\pi/2T)$, $u_j = (\pi/4)(1 + \cos((2j-1)\pi/2T))$, T is an accuracy and complexity tradeoff parameter, and the antenna gain $G_L(\cdot, \cdot)$ is given by

$$G_L(x, y) = \frac{\sin^2\left(\frac{\pi M(f_c D(\cos(x) - \cos(\theta_b)) - (y - d_{ab})\Delta f)}{c}\right)}{M \sin^2\left(\frac{\pi(f_c D(\cos(x) - \cos(\theta_b)) - (y - d_{ab})\Delta f)}{c}\right)}. \quad (37)$$

Proof: Refer to Appendix C. ■

Remark 3: By comparing (21) and (35), we find that the covertness performance difference between the two beamforming schemes is mainly determined by the average channel gain at the most malicious Willie, i.e., Ω_P and Ω_L . Specifically, the smaller $\Omega_P(\Omega_L)$ means the weaker observation samples at the most malicious Willie and thus the better covertness performance. We recall that, in Remark 2, the effect of θ_b

on Ω_P is characterized via analyzing the variation of beamwidth. This result helps to analyze the covertness performance while communicating with the receivers at different orientations for the PA beamforming scheme. However, under the LFDA beamforming scheme, the beam pattern depends on both angle and distance, which cannot intuitively reflect the effect on Ω_L . Therefore, we analyze the effect of θ_b on Ω_L via simulating (36) versus θ_b in Fig. 3, and the Monte Carlo simulations are presented to verify the accuracy of the derivations. It can be observed that Ω_P first decreases and then increases as θ_b increases from 0° to 180° , which verifies the analysis in Remark 2. Besides, Fig. 3 shows that the outperformance between Ω_L and Ω_P depends on the orientation of Bob. This means that Alice can properly adopt different beamforming schemes to obtain superior covert communication performance, in which the proposed theoretical analysis can provide guidelines for choosing the PA or LFDA beamforming scheme.

B. Optimization of Average Effective Covert Throughput

As discussed above, we use the AECT to measure the performance of covert communications. Similar to (25), the AECT for the LFDA beamforming scheme can be written as

$$\bar{\eta}_L = NR(1 - \mathbb{E}(\delta)), \text{ s.t. (35)} \quad (38)$$

where the expectation $\mathbb{E}(\delta)$ is given by (27).

Accordingly, the optimization problem to maximize the AECT for the LFDA beamforming scheme by considering the maximum available blocklength constraint can be written as

$$\begin{aligned} \max_{N, P_a} \quad & \bar{\eta}_L, \\ \text{s.t.} \quad & (35), N \leq N_{\max}, N \in \mathbb{N}^+. \end{aligned} \quad (39)$$

Similar to (31), the optimal transmit power $P_{a,L}^+$ for a given N can be written as

$$P_{a,L}^+ = \frac{2\varepsilon\sigma^2}{\sqrt{N}\Omega_L} \quad (40)$$

where Ω_L is given by (36). Then, the optimization problem (39) is tackled by resorting to the following theorem.

Theorem 4: Under the LFDA beamforming scheme, the optimal transmit power and blocklength maximizing the AECT, i.e., the solution of (39), can be given by

$$P_{a,L}^* = \frac{2\varepsilon\sigma^2}{\sqrt{N_L^*}\Omega_L} \quad (41)$$

$$\begin{aligned} \Omega_L = & \frac{\lambda_w \pi^3}{T^3} \sum_{k=1}^T \sum_{i=1}^T \sum_{j=1}^T \frac{e^{-\frac{\tan^\alpha(z_j) \tan(u_k)}{G_L(v_i, \tan(z_j))}} \tan(u_k) \tan^{1+\alpha}(z_j)}{G_L(v_i, \tan(z_j)) \cos^2(u_k) \cos^2(z_j)} \sqrt{v_i(2\pi - v_i) \left(z_j - \frac{\pi}{4}\right) \left(\frac{\pi}{2} - z_j\right) u_k \left(\frac{\pi}{2} - u_k\right)} \\ & \times \exp\left(-\frac{\lambda_w \pi^2}{T^2} \sum_{i=1}^T \sum_{j=1}^T \frac{\tan(z_j) e^{-\frac{\tan^\alpha(z_j) \tan(u_k)}{G_L(v_i, \tan(z_j))}}}{\cos^2(z_j)} \sqrt{\left(z_j - \frac{\pi}{4}\right) \left(\frac{\pi}{2} - z_j\right) v_i(2\pi - v_i)}\right). \end{aligned} \quad (36)$$

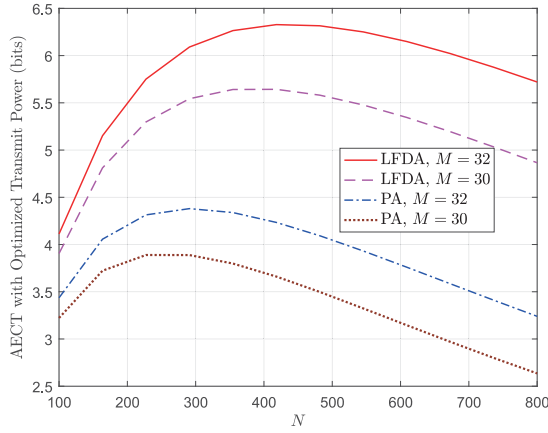


Fig. 4. AECT with the optimized transmit power versus the blocklength, where $d_{ab} = 85$ m, $\theta_b = 45^\circ$, $\lambda_w = 3 \times 10^{-5}$, $\sigma^2 = 0$ dBm, and $\varepsilon = 0.1$.

and

$$N_L^* = \begin{cases} \left\lfloor \left(\frac{\beta_b}{2\gamma_0} \right)^2 \right\rfloor, & N_L^0 \leq \left(\frac{\beta_b}{2\gamma_0} \right)^2 \\ \min(N_{\max}, \lfloor N_L^0 \rfloor), & N_L^0 > \left(\frac{\beta_b}{2\gamma_0} \right)^2 \end{cases} \quad (42)$$

where $N_L^0 = (4M\varepsilon/(\gamma_0\Omega_L d_{ab}^\alpha))^2$.

Proof: The proof is similar to the proof for Theorem 2 but replacing Ω_P with Ω_L . ■

Remark 4: From both (31) and (40), we can observe that the optimized transmit power is a decreasing function of N for both beamforming schemes. Meanwhile, it is known that the AECT is an increasing function versus P_a and N , which demonstrates that there exists a tradeoff between P_a and N to maximize the AECT. Besides, according to (33) and (42), the optimal blocklength may not equal to the maximal available blocklength $N_{\max}$, and thus the maximum AECT cannot always be improved by increasing $N_{\max}$. Moreover, from Theorems 2 and 4, we find that the optimal transmit power and blocklength decrease as the average channel gain at the most malicious Willie $\Omega_P(\Omega_L)$ increases. Since that the AECT becomes small as the transmit power and blocklength decrease, the performance difference of covert communication for the PA and LFDA beamforming schemes can be reflected on the variation of $\Omega_P(\Omega_L)$, and the larger $\Omega_P(\Omega_L)$ means the smaller AECT and the worse covert communication performance.

V. NUMERICAL RESULTS

In this section, the numerical results are presented by considering both the PA and LFDA beamforming schemes. In the simulations, the channel coding rate is set as $R = 0.1$ bits per channel use (bpcu), the carrier frequency is set as $f_c = 28$ GHz, the path-loss exponent at 28 GHz is set as $\alpha = 2.2$ [47], the antenna interval is set as $D = c/(2f_c)$, and the frequency increment for the LFDA beamforming scheme is set as $\Delta f = 1$ MHz.

Fig. 4 shows the AECT with the optimized transmit power versus the blocklength N for different numbers of antennas M . The optimized transmit power for the PA and LFDA beamforming schemes are given by (31) and (40), respectively. First,

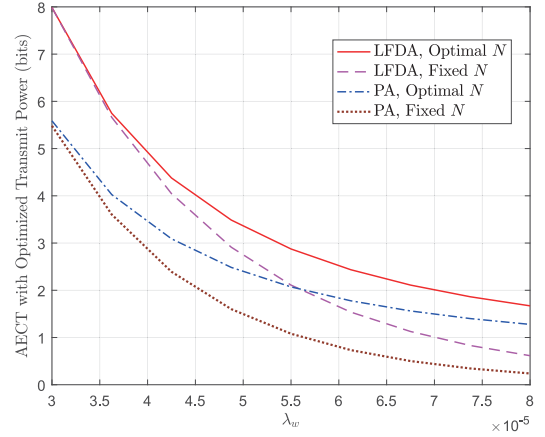


Fig. 5. AECT with the optimized transmit power versus Willie's density, where $d_{ab} = 80$ m, $\theta_b = 45^\circ$, $N_{\max} = 500$, $M = 32$, $\sigma^2 = 0$ dBm, and $\varepsilon = 0.1$.

it can be observed that the AECT with the optimized transmit power initially increases and then decreases as N becomes large for both of the PA and LFDA beamforming schemes. This is because that the AECT is influenced by the tradeoff between the transmit power and blocklength. For the small value of N , the AECT is mainly constrained by the blocklength, and the improvement by increasing N exceeds the performance reduction caused by the decreasing of transmit power. For a large value of N , the transmit power should be small due to the covert communication constraint, and thus the AECT is mainly constrained by the transmit power. Besides, we observe that the AECT with optimized transmit power is improved when M changes from 30 to 32 for both beamforming schemes. It can be explained by the fact that the received signals at Bob can be enhanced due to the larger antenna gain while the received signals at the most malicious Willie will not always be reinforced due to its random location.

Fig. 5 depicts the AECT with optimized transmit power versus Willie's density λ_w under both the cases with optimal blocklength and fixed blocklength ($N = N_{\max}$). First, it can be observed that the curves of the AECT with optimized transmit power decreases as λ_w increases for both the PA and LFDA beamforming schemes. This is due to that larger λ_w means that there will be more Willies performing the detection tasks, and thus it is more likely to obtain the stronger detection channels for Willies. By comparing the optimized AECT between the cases with the fixed blocklength and the optimal blocklength, we can observe that the larger AECT can be achieved by adopting the optimal blocklength, which illustrates the effectiveness of our proposed transmission design for enhancing the covert communication performance. Also, the results show that the improvement of adopting the optimal blocklength becomes more significant when λ_w increases. The reason is that, when λ_w is small, the optimal N is constrained by $N_{\max}$. That is, $N_{\max}$ locates at the left side of the optimal N in Fig. 4, and thus the optimal value of N equals to $N_{\max}$. For the large value of λ_w , the optimal value of N is smaller than $N_{\max}$, and the difference between the optimal N and $N_{\max}$ enlarges when λ_w increases. This observation reveals that it is more necessary

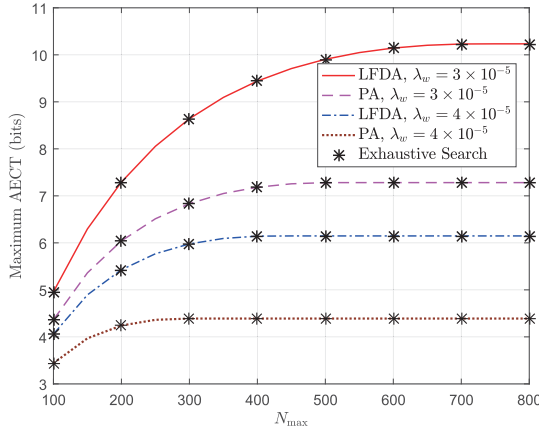


Fig. 6. Maximum AECT versus the maximal available blocklength, where $d_{ab} = 75$ m, $\theta_b = 45^\circ$, $M = 32$, $\sigma^2 = 0$ dBm, and $\varepsilon = 0.1$.

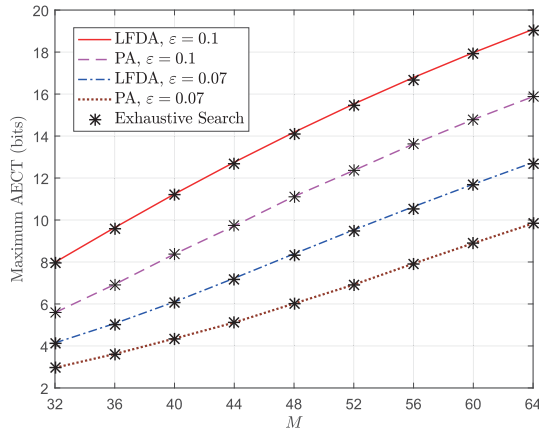


Fig. 7. Maximum AECT versus the number of antennas, where $d_{ab} = 80$ m, $\theta_b = 45^\circ$, $N_{\max} = 500$, $\sigma^2 = 0$ dBm, and $\lambda_w = 3 \times 10^{-5}$.

and also more effective to optimize the blocklength when λ_w is large as compared to the case with small λ_w .

Fig. 6 plots the maximum AECT versus the maximal available blocklength $N_{\max}$ for different Willie's densities λ_w . The analysis results are obtained by adopting the optimal transmit power and blocklength, which are given by Theorem 2 and Theorem 4. First, it can be observed that the analysis results agree with the exhaustive search results well, which verifies the accuracy of the derivations. Besides, it is shown that the maximum AECT increases and then remains constant as a function of $N_{\max}$ for both the PA and LFDA beamforming schemes. This observation can be explained by the fact that, in the small $N_{\max}$ regime, the maximum AECT mainly constrained by the value of $N_{\max}$, and thus can be improved by enlarging $N_{\max}$. However, when $N_{\max}$ is large, the optimal N is smaller than $N_{\max}$, and thus increasing $N_{\max}$ will not influence the maximum AECT. Moreover, Fig. 6 shows that, for both the beamforming schemes, the maximum AECT with $\lambda_w = 4 \times 10^{-5}$ is smaller than that with $\lambda_w = 3 \times 10^{-5}$ due to the larger number of Willies.

In Fig. 7, we plot the maximum AECT versus the number of antennas M for different covertness tolerances ε . First, we can observe that the maximum AECT increases as M becomes

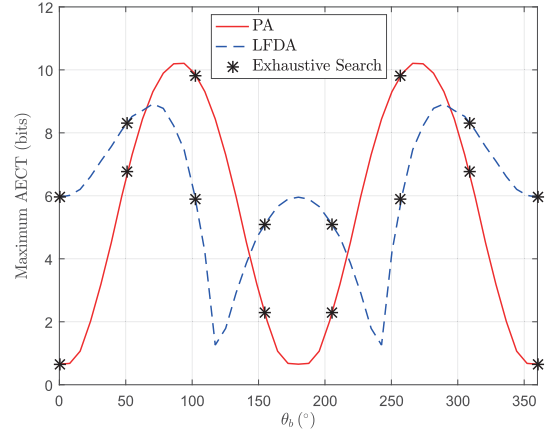


Fig. 8. Maximum AECT versus the direction of Bob, where $d_{ab} = 80$ m, $N_{\max} = 500$, $M = 32$, $\lambda_w = 3 \times 10^{-5}$, $\sigma^2 = 0$ dBm, and $\varepsilon = 0.1$.

large due to the stronger received signals at Bob, which is also shown in Fig. 4. Moreover, we find that, for both the PA and LFDA beamforming schemes, the maximum AECT enlarges as ε changes from 0.07 to 0.1. This is due to that the larger ε relaxes the covert communication constraint, and thus Alice can adopt the larger blocklength and transmit power to increase the maximum AECT.

In Fig. 8, we plot the maximum AECT versus the direction of Bob θ_b for both the PA and LFDA beamforming schemes. First, we can observe that the curves are symmetric about the axis of $\theta_b = 180^\circ$ due to the symmetry of the ULA. Besides, the results also show that, for the PA beamforming scheme, the maximum AECT first increases and then decreases as θ_b increases from 0° to 180° . This is because that the beam-width initially increases and then decreases as the orientation of beam changes from 0° to 180° (the reasons are detailed in Remark 2). Then, in the large beam-width regime, Willies have more chances to obtain the high antenna gain and thus the stronger received signals. Accordingly, when the beam is narrow, it is difficult for Willie to obtain the effective observations. On the other hand, the beam pattern for the LFDA beamforming scheme is different from that for the PA beamforming scheme due to the existence of frequency increment. In this regard, it can be observed that whether the PA or LFDA beamforming scheme is superior relates with the direction of Bob. This indicates that the transmitter can adaptively choose the PA or LFDA beamforming scheme according to the orientation of the legitimate receiver for the purpose of enhancing the covert communication performance.

VI. CONCLUSION

In this article, we investigated the covert mmWave communication in the finite blocklength regime, while considering spatially random Willies trying to detect whether Alice transmits or not. For both the PA and LFDA beamforming schemes, the novel expressions of covert communication constraint were derived by using the tools of stochastic geometry. To facilitate the covert transmission design, the optimal transmit power and blocklength were derived to maximize the AECT for both beamforming schemes, which demonstrated that there exists a tradeoff between the transmit power and blocklength

to maximize the AECT. Setting the fixed blocklength case as a benchmark, it was revealed that the enhancement of AECT by utilizing the optimal blocklength enlarges as Willie's density increases. Besides, for both the beamforming schemes, the performance of covert communications may not be improved by increasing the maximal available blocklength, but it can be indeed enhanced by increasing the number of antennas. Furthermore, the numerical results showed that the transmitter can adaptively choose the PA or LFDA scheme according to the orientation of user to improve the covert mmWave communication performance against spatially random Willies.

APPENDIX A PROOF OF THEOREM 1

Substituting (19) into (20), the covert communication constraint can be written as

$$\mathbb{E} \left(\max_{w_l \in \Phi_w} \sqrt{\frac{N}{2}} \left(\frac{P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2}{\sigma^2} - \ln \left(1 + \frac{P_a |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2}{\sigma^2} \right) \right)^{\frac{1}{2}} \right) \leq \varepsilon. \quad (43)$$

As suggested by [48], the received SNR is usually low in covert communications, and thus (43) can be further rewritten by using the bound $\ln(1+x) \geq x - x^2/2$, $x > 0$ [49], i.e.,

$$\frac{P_a \sqrt{N}}{2\sigma^2} \mathbb{E} \left(\underbrace{\max_{w_l \in \Phi_w} |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2}_{\kappa_1} \right) \leq \varepsilon \quad (44)$$

where $|\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2$ can be derived as

$$|\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2 = \frac{|a_{w_l}|^2}{d_{aw_l}^\alpha} \left| \frac{1}{\sqrt{M}} \mathbf{a}_P(\theta_{w_l}) \mathbf{a}_P^H(\theta_b) \right|^2$$

$$\begin{aligned} &= \frac{|a_{w_l}|^2}{d_{aw_l}^\alpha M} \left| \sum_{m=0}^{M-1} e^{j \left(m - \frac{M-1}{2} \right) \frac{2\pi D f_c}{c} (\cos(\theta_{w_l}) - \cos(\theta_b))} \right|^2 \\ &\stackrel{(a)}{=} \frac{|a_{w_l}|^2}{d_{aw_l}^\alpha} \frac{\sin^2 \left(\frac{M\pi D f_c}{c} (\cos(\theta_{w_l}) - \cos(\theta_b)) \right)}{M \sin^2 \left(\frac{\pi D f_c}{c} (\cos(\theta_{w_l}) - \cos(\theta_b)) \right)} \end{aligned} \quad (45)$$

and step (a) is due to the sum of geometric progression.

Then, similar to (47), the CDF of κ_2 can be derived as

$$\begin{aligned} F_{\kappa_2}(t) &= \Pr \left(\max_{w_l \in \Phi_w} |\mathbf{h}_{L,aw_l} \mathbf{w}_L|^2 \leq t \right) \\ &= \exp \left(-\lambda_w \int_0^{2\pi} \int_1^\infty \Pr \left(|a_{w_l}|^2 \geq \frac{r^\alpha t}{G_L(\theta, r)} \right) r dr d\theta \right) \\ &= \exp \left(-\lambda_w \int_0^{2\pi} \int_1^\infty r e^{-\frac{r^\alpha t}{G_L(\theta, r)}} dr d\theta \right) \end{aligned} \quad (46)$$

where $G_L(\cdot, \cdot)$ denotes the antenna gain under the LFDA beamforming scheme, which is expressed as (37).

In order to derive the expectation of κ_1 , we first derive the CDF of κ_1 as follows:

$$\begin{aligned} F_{\kappa_1}(t) &= \Pr \left(\max_{w_l \in \Phi_w} |\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2 \leq t \right) \\ &= \Pr \left(\min_{w_l \in \Phi_w} \frac{1}{|\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2} \geq \frac{1}{t} \right) \\ &\stackrel{(a)}{=} \exp \left(-\lambda_w \int_{\mathbb{R}^2} \Pr \left(\frac{1}{|\mathbf{h}_{P,aw_l} \mathbf{w}_P|^2} \in \left[0, \frac{1}{t} \right] \right) dw_l \right) \\ &= \exp \left(-\lambda_w \int_0^{2\pi} \int_1^\infty \Pr \left(|a_{w_l}|^2 \geq \frac{r^\alpha t}{G_P(\theta)} \right) r dr d\theta \right) \\ &= \exp \left(-\lambda_w \int_0^{2\pi} \int_1^\infty r e^{-\frac{r^\alpha t}{G_P(\theta)}} dr d\theta \right) \\ &\stackrel{(b)}{=} \exp \left(-\frac{\lambda_w t^{-\frac{2}{\alpha}}}{\alpha} \int_0^{2\pi} \Gamma \left(\frac{2}{\alpha}, \frac{t}{G_P(\theta)} \right) (G_P(\theta))^{\frac{2}{\alpha}} d\theta \right) \end{aligned} \quad (47)$$

$$\begin{aligned} \mathbb{E}(\kappa_1) &= \int_0^\infty t f_{\kappa_1}(t) dt \\ &= \int_0^\infty \exp \left(-\frac{\lambda_w t^{-\frac{2}{\alpha}}}{\alpha} \int_0^{2\pi} \Gamma \left(\frac{2}{\alpha}, \frac{t}{G_P(\theta)} \right) (G_P(\theta))^{\frac{2}{\alpha}} d\theta \right) \left(\frac{\lambda_w}{\alpha} \int_0^{2\pi} e^{-\frac{t}{G_P(\theta)}} d\theta \right. \\ &\quad \left. + \frac{2\lambda_w t^{-\frac{2}{\alpha}}}{\alpha^2} \int_0^{2\pi} \Gamma \left(\frac{2}{\alpha}, \frac{t}{G_P(\theta)} \right) (G_P(\theta))^{\frac{2}{\alpha}} d\theta \right) dt \\ &\stackrel{(a)}{=} \int_0^{\frac{\pi}{2}} \exp \left(-\frac{\lambda_w \tan^{-\frac{2}{\alpha}}(u)}{\alpha} \int_0^{2\pi} \Gamma \left(\frac{2}{\alpha}, \frac{\tan(u)}{G_P(\theta)} \right) (G_P(\theta))^{\frac{2}{\alpha}} d\theta \right) \left(\frac{\lambda_w}{\alpha} \int_0^{2\pi} e^{-\frac{\tan(u)}{G_P(\theta)}} d\theta \right. \\ &\quad \left. + \frac{2\lambda_w \tan^{-\frac{2}{\alpha}}(u)}{\alpha^2} \int_0^{2\pi} \Gamma \left(\frac{2}{\alpha}, \frac{\tan(u)}{G_P(\theta)} \right) (G_P(\theta))^{\frac{2}{\alpha}} d\theta \right) \frac{du}{\cos^2(u)} \\ &\stackrel{(b)}{=} \frac{\pi \lambda_w}{T} \sum_{j=1}^T \exp \left(-\frac{\lambda_w \tan^{-\frac{2}{\alpha}}(u_j) \Xi_P(u_j)}{\alpha} \right) \left(\frac{\pi}{T\alpha} \sum_{i=1}^T e^{-\frac{\tan(u_j)}{G_P(v_i)}} \sqrt{v_i(2\pi - v_i)} + \frac{2 \tan^{-\frac{2}{\alpha}}(u_j) \Xi_P(u_j)}{\alpha^2} \right) \frac{\sqrt{u_j(\frac{\pi}{2} - u_j)}}{\cos^2(u_j)}. \end{aligned} \quad (48)$$

where step (a) is due to the void probability of PPP [50], step (b) is obtained by using [45, eq. (2.33.10)], and $G_P(\cdot)$ denotes the antenna gain for the PA beamforming scheme, which is given by (23).

Then, the PDF of κ_1 can be calculated by $f_{\kappa_1}(t) = dF_{\kappa_1}(t)/dt$, and the expectation of κ_1 can be derived as (48), shown at the bottom of the previous page, where $\Xi_P(\cdot)$ is given by (24), step (a) is due to the substitution $t = \tan(u)$, step (b) is obtained by using the Gaussian–Chebyshev quadrature [51], [52], $v_i = \pi(1 + \cos([(2i-1)\pi]/2T))$, $u_j = (\pi/4)(1 + \cos([(2j-1)\pi]/2T))$, and T is an accuracy and complexity tradeoff parameter.

Finally, substituting (48) into (44), the proof is completed.

APPENDIX B PROOF OF THEOREM 2

Substituting (31) into $\bar{\eta}_P$, the optimization problem (30) can be converted into 1-D problem, and then we can derive the optimal blocklength as below.

According to (29), the CDF of γ_b is piecewise, and we consider two cases of $N \leq (\beta_b/(2\gamma_0))^2$ and $N > (\beta_b/[2\gamma_0])^2$.

Case 1: When $N \leq (\beta_b/[2\gamma_0])^2$, the AECT with $P_a = P_{a,P}^+$ can be expressed

$$\begin{aligned}\bar{\eta}_P^+ &= NR \left(\frac{P_{a,P}^+ M \sqrt{N}}{\sigma^2 d_{ab}^\alpha \beta_b} F_{\gamma_b} \left(\gamma_0 + \frac{\beta_b}{2\sqrt{N}} \right) \right) \\ &= NR \frac{2\epsilon M}{\Omega_P d_{ab}^\alpha \beta_b} \left(1 - e^{-\frac{(\sqrt{N}\gamma_0 + \frac{\beta_b}{2}) d_{ab}^\alpha \Omega_P}{2\epsilon M}} \right)\end{aligned}\quad (49)$$

where the CDF $F_{\gamma_b}(\cdot)$ is given by (29).

Then, the first derivative for $\bar{\eta}_P^+$ versus N can be derived as

$$\begin{aligned}\frac{\partial \bar{\eta}_P^+}{\partial N} &= \frac{2\epsilon RM}{\Omega_P d_{ab}^\alpha \beta_b} \left(1 - e^{-\frac{(\sqrt{N}\gamma_0 + \frac{\beta_b}{2}) d_{ab}^\alpha \Omega_P}{2\epsilon M}} \right) \\ &\quad \times \left(1 - \frac{\sqrt{N}\gamma_0 d_{ab}^\alpha \Omega_P}{4\epsilon M} \right).\end{aligned}\quad (50)$$

It is observed that $\bar{\eta}_P^+$ increases as N becomes large due to $(\partial \bar{\eta}_P^+ / \partial N) \geq 0$. Then, considering that N is a positive integer, the optimal blocklength can be given by $\lfloor (\beta_b/(2\gamma_0))^2 \rfloor$ for case 1.

Case 2: When $N > (\beta_b/[2\gamma_0])^2$, the AECT with $P = P_{a,P}^+$ is given by

$$\begin{aligned}\bar{\eta}_P^+ &= NR \frac{P_{a,P}^+ M \sqrt{N}}{\sigma^2 d_{ab}^\alpha \beta_b} \left(F_{\gamma_b} \left(\gamma_0 + \frac{\beta_b}{2\sqrt{N}} \right) - F_{\gamma_b} \left(\gamma_0 - \frac{\beta_b}{2\sqrt{N}} \right) \right) \\ &= \frac{2NR\epsilon M}{\Omega_P d_{ab}^\alpha \beta_b} \left(e^{-\frac{(\sqrt{N}\gamma_0 - \frac{\beta_b}{2}) \Omega_P d_{ab}^\alpha}{2\epsilon M}} - e^{-\frac{(\sqrt{N}\gamma_0 + \frac{\beta_b}{2}) \Omega_P d_{ab}^\alpha}{2\epsilon M}} \right) \\ &= Ne^{-\frac{\sqrt{N}\gamma_0 \Omega_P d_{ab}^\alpha}{2\epsilon M}} \frac{2\epsilon RM}{\Omega_P d_{ab}^\alpha \beta_b} \left(e^{\frac{\Omega_P d_{ab}^\alpha \beta_b}{4\epsilon M}} - e^{-\frac{\Omega_P d_{ab}^\alpha \beta_b}{4\epsilon M}} \right).\end{aligned}\quad (51)$$

For simplicity, we denote the right two terms of (51), which do not contain N , as C . Then, the first derivative of $\bar{\eta}_P^+$ versus N can be derived as

$$\frac{\partial \bar{\eta}_P^+}{\partial N} = Ce^{-\frac{\sqrt{N}\gamma_0 \Omega_P d_{ab}^\alpha}{2\epsilon M}} \left(1 - \frac{\sqrt{N}\gamma_0 \Omega_P d_{ab}^\alpha}{4\epsilon M} \right).\quad (52)$$

Based on (52), we can obtain the extreme point N_P^0 , of which the expression can be given by

$$N_P^0 = \left(\frac{4\epsilon M}{\gamma_0 \Omega_P d_{ab}^\alpha} \right)^2.\quad (53)$$

$$\begin{aligned}|\mathbf{h}_{L,aw_l} \mathbf{w}_L|^2 &= \frac{|a_{w_l}|^2}{d_{aw_l}^\alpha} \left| \frac{1}{\sqrt{M}} \mathbf{a}_L(\theta_{w_l}, d_{aw_l}) \mathbf{a}_L^H(\theta_b, d_{ab}) \right|^2 \\ &= \frac{|a_{w_l}|^2}{M d_{aw_l}^\alpha} \sum_{m=0}^{M-1} \sum_{m'=0}^{M-1} e^{-j\frac{2\pi}{c} (f_c(m - \frac{M-1}{2}) D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab})(m - \frac{M-1}{2}) \Delta f)} \\ &\quad \times e^{j\frac{2\pi}{c} (f_c(m' - \frac{M-1}{2}) D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab})(m' - \frac{M-1}{2}) \Delta f)} \\ &= \frac{|a_{w_l}|^2}{M d_{aw_l}^\alpha} \sum_{m=0}^{M-1} \sum_{m'=0}^{M-1} e^{-j\frac{2\pi m}{c} (f_c D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab}) \Delta f)} e^{j\frac{2\pi m'}{c} (f_c D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab}) \Delta f)} \\ &\stackrel{(a)}{=} \frac{|a_{w_l}|^2}{M d_{aw_l}^\alpha} \left(\frac{\sin\left(\frac{\pi M (f_c D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab}) \Delta f)}{c}\right)}{\sin\left(\frac{\pi (f_c D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab}) \Delta f)}{c}\right)} \right) \frac{e^{-j\frac{\pi M (f_c D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab}) \Delta f)}{c}}}{e^{-j\frac{\pi (f_c D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab}) \Delta f)}{c}}} \\ &\quad \times \sum_{m'=0}^{M-1} e^{j\frac{2\pi m'}{c} (f_c D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab}) \Delta f)} \\ &= \frac{|a_{w_l}|^2}{d_{aw_l}^\alpha} \frac{\sin^2\left(\frac{\pi M (f_c D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab}) \Delta f)}{c}\right)}{M \sin^2\left(\frac{\pi (f_c D(\cos(\theta_{w_l}) - \cos(\theta_b)) - (d_{aw_l} - d_{ab}) \Delta f)}{c}\right)}.\end{aligned}\quad (56)$$

$$\begin{aligned}
\mathbb{E}(\kappa_2) &= \int_0^\infty t f_{\kappa_2}(t) dt \\
&= \lambda_w \int_0^\infty \int_0^{2\pi} \int_1^\infty \frac{r^{1+\alpha} t}{G_L(\theta, r)} e^{-\frac{r^\alpha t}{G_L(\theta, r)}} dr d\theta \exp\left(-\lambda_w \int_0^{2\pi} \int_1^\infty r e^{-\frac{r^\alpha t}{G_L(\theta, r)}} dr d\theta\right) dt \\
&\stackrel{a}{=} \lambda_w \int_0^{\frac{\pi}{2}} \int_0^{2\pi} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{e^{-\frac{\tan^\alpha(z) \tan(u)}{G_L(\theta, \tan(z))}} \tan^{1+\alpha}(z) \tan(u)}{G_L(\theta, \tan(z)) \cos^2(z) \cos^2(u)} dz d\theta \exp\left(-\lambda_w \int_0^{2\pi} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\tan(z) e^{-\frac{\tan^\alpha(z) \tan(u)}{G_L(\theta, \tan(z))}}}{\cos^2(z)} dz d\theta\right) du \\
&\stackrel{b}{=} \frac{\lambda_w \pi^3}{T^3} \sum_{k=1}^T \sum_{i=1}^T \sum_{j=1}^T \frac{e^{-\frac{\tan^\alpha(z_j) \tan(u_k)}{G_L(v_i, \tan(z_j))}} \tan(u_k) \tan^{1+\alpha}(z_j)}{G_L(v_i, \tan(z_j)) \cos^2(u_k) \cos^2(z_j)} \sqrt{v_i(2\pi - v_i) \left(z_j - \frac{\pi}{4}\right) \left(\frac{\pi}{2} - z_j\right) u_k \left(\frac{\pi}{2} - u_k\right)} \\
&\quad \times \exp\left(-\frac{\lambda_w \pi^2}{T^2} \sum_i \sum_j \frac{\tan(z_j) e^{-\frac{\tan^\alpha(z_j) \tan(u_k)}{G_L(v_i, \tan(z_j))}}}{\cos^2(z_j)} \sqrt{\left(z_j - \frac{\pi}{4}\right) \left(\frac{\pi}{2} - z_j\right) v_i(2\pi - v_i)}\right). \tag{58}
\end{aligned}$$

Next, we derive the second derivative of $\bar{\eta}_p^+$ subject to N , i.e.,

$$\begin{aligned}
\frac{\partial^2 \bar{\eta}_p^+}{\partial N^2} &= C \left(e^{-\frac{\sqrt{N} \gamma_0 \Omega p d_{ab}^\alpha}{2M\varepsilon}} \left(-\frac{\gamma_0 \Omega p d_{ab}^\alpha}{4\sqrt{NM}\varepsilon} \right) \left(1 - \frac{\sqrt{N} \gamma_0 \Omega p d_{ab}^\alpha}{4M\varepsilon} \right) \right. \\
&\quad \left. + e^{-\frac{\sqrt{N} \gamma_0 \Omega p d_{ab}^\alpha}{2M\varepsilon}} \left(-\frac{\gamma_0 \Omega p d_{ab}^\alpha}{8\sqrt{NM}\varepsilon} \right) \right) \\
&= -\frac{\gamma_0 \Omega p d_{ab}^\alpha}{4\sqrt{NM}\varepsilon} \frac{\partial \bar{\eta}_p^+}{\partial N} - \frac{C \gamma_0 \Omega p d_{ab}^\alpha}{8\sqrt{NM}\varepsilon} e^{-\frac{\sqrt{N} \gamma_0 \Omega p d_{ab}^\alpha}{2\varepsilon}}. \tag{54}
\end{aligned}$$

As we can see, when $N = N_p^0$, we have $(\partial^2 \bar{\eta}_p^+ / \partial N^2)|_{N=N_p^0} < 0$. This means that $\bar{\eta}_p^+$ is quasi-concave on N [28], i.e., $\bar{\eta}_p^+$ first increases and then decreases versus N and is maximized at $N = N_p^0$. Accordingly, recalling that N is a positive integer and limited by $N \leq N_{\max}$, we can obtain the optimal blocklength $N_p^* = \min(N_{\max}, \lfloor N_p^0 \rfloor)$ for Case 2.

Finally, combining cases 1 and 2, and substituting the result N_p^* into (31), the proof is completed.

APPENDIX C PROOF OF THEOREM 3

Similar to (44), by using the inequality $\ln(1+x) \geq x - x^2/2$ for any $x > 0$, the covert communication constraint for the LFDA beamforming scheme is further rewritten as

$$\frac{P_a \sqrt{N}}{2\sigma^2} \mathbb{E} \left(\underbrace{\max_{w_l \in \Phi_w} |\mathbf{h}_{L, awl} \mathbf{w}_L|^2}_{\kappa_2} \right) \leq \varepsilon \tag{57}$$

where $|\mathbf{h}_{L, awl} \mathbf{w}_L|^2$ can be derived as (56) shown at the bottom of the previous page, where step (a) follows the sum of geometric progression.

Based on (46), the PDF of κ_2 can be obtained by $f_{\kappa_2}(t) = dF_{\kappa_2}(t)/dt$. Thus, the expectation $\mathbb{E}(\kappa_2)$ can be derived as (58), shown at the top of the page, where step (a) is due to the substitutions $t = \tan(u)$ and $r = \tan(z)$, step (b) follows the Gaussian–Chebyshev quadrature, $v_i = \pi(1 + \cos[(2i-1)\pi]/2T)$, $z_j = (3\pi/8) + (\pi/8) \cos[(2j-1)$

$\pi]/2T)$, $u_j = (\pi/4)(1 + \cos[(2j-1)\pi]/2T)$, and T is an accuracy and complexity tradeoff parameter.

Finally, substituting (58) into (57), the proof is completed.

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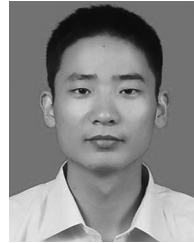
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